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All-optical variable-length packet router with contention resolution based on wavelength conversion

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ABSTRACT

We proposed a novel 2×2 all optical packet switching router architecture supporting asynchronous variable-length packet. In order to deal with the contention problem we adopt for wavelength conversion strategy. A proof of concept through Optiwave simulation is validated. We have showing that the contending packet is detected and forwarded according FIFO (First In First Out) strategy at a different wavelength. Error-free functionality is achieved for high bit rates (up to 100 Gbps).

Keywords: packet, forwarding, contention; resolution; wavelength conversion; all-optical flip-flop; optical AND gate; SOA-MZI.

1. INTRODUCTION

As the data rate grows year on year, the demand for network speed and capacity. Data rate per fiber at the backbone network reaches the order of Tbits/s based on wavelength division and multiplexing (WDM), Optical Code Division Multiple Access (OCDMA) etc. Thus ultra fast speed systems are required for next generation access networks. The electrical routers face the huge challenge of how to satisfy the switching requirement of explosive data. The processing and switching speed is limited by the inherent parasitic parameters in electrical routers and the electronic weakness appears. The idea of replacing electrical switching with optical switching attracts lots of attention. All-optical switches are expected to play an important role and replace conventional electronic or electro-optical switches to take advantage of their high integration capabilities, their reduced cost and low power consumption. Several photonic switching subsystems are developed in literature [1-3].

All optical logic gates are very important element in the realization of different optical networks tasks like header recognition, packet synchronization, switching, signal regeneration, and signal processing. For the design of those optical logic gates, a nonlinear medium such as SOA is required in order to modulate the signal and generate the desired outputs [4]. The optical logical AND gate is the most used. In fact it is indispensable to critical for optical network nodes. It is a key element for optical signal switching [3-7]. Optical flip flop is an essential unit too for optical signal processing [7-9]. Different technologies are used for the implementation of the flip flop and the AND gate. The most used is the Semiconductor Optical Amplifier based Mach–Zehnder Interferometer (SOA_MZI) [3-9].

In OPS routers two concepts must be considered that is the packets routing and the contention resolution. The principal of packed routing is based on searching and reaching the final destination. So each intermediate node needs to make a decision to which next node send the incoming packet. Contention occurs when two or more packets arrive simultaneously from different inputs of a network node and contend for the same output port. However, only one packet can be switched to any given output at once. Thus, several techniques are used to detect and resolve contention in all optical networks. As a result, losing contending packet is avoided. These contention resolution techniques can be classified into three categories that are time buffering, deflection routing and wavelength conversion. The optical buffer stores the contending packet. When the requested output port is empty this packet will be released [10]. Nevertheless, deflection routing consists on resort to space dimension by sending the contending packet to a second preferred output port [11, 12]. However, wave conversion technique converts the contending packet to another wavelength [13, 14].

This manuscript is organized as follows. Section II describes the router architecture based on wavelength conversion for contention resolution. Section III presents and analyzes results and discussions, while the final section provides conclusions.

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2. ALL OPTICAL PACKET ROUTER ARCHITECTURE

2.1 Packet forwarding

In [7] we demonstrate an all optical packet forwarding architecture. The forwarding function involves two steps: generating an ON/OFF state with duration of the data payload, then performing the AND logical operation on the packet and the flip-flop output. We note here that these two steps are achieved separately although both of them utilize SOA-MZI based elements.

The packet could have a different length with a variable bit rate. Therefore signals indicating its end and beginning are required. As described in Fig.1.(a), packet frames are constructed by inducing a set signal in front of the payload and a reset signal at its end. A guard time is introduced between the packet‘s signaling parts and the payload part to facilitate their extraction.

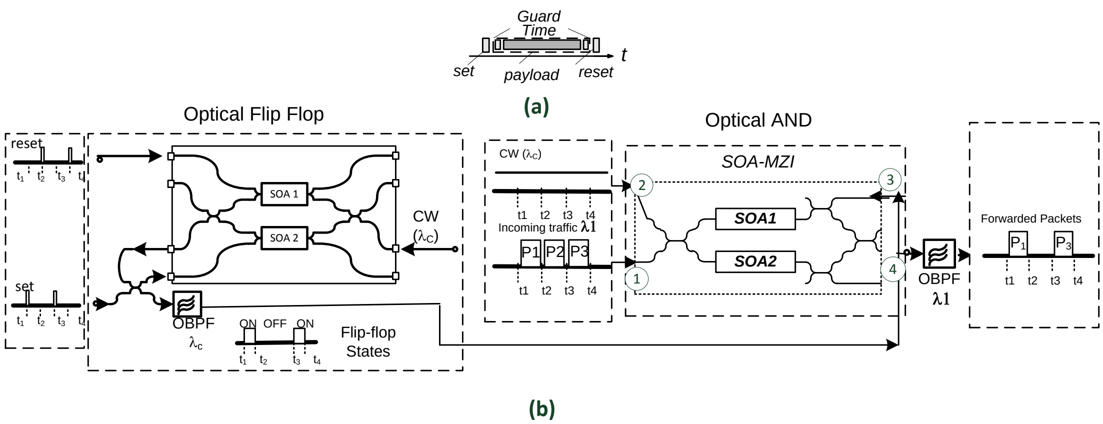

Figure 1:(a) packet structure, (b) all optical packet forwarding architecture [7].

As described in Fig.1.(b) the packet forwarding unit employs an all optical flip-flop and an optical AND gate that are based on SOA _MZIs. While the Set and Reset pulses effectively switched the output states of the flip-flop, the ON state serves as the suitable control signal to obtain the desirable logical AND operation with the forwarded packet P1 and P3 as described in the example on Fig.1.(b). When receiving the set pulse the flip flop start and continue sending a logical pulse thanks to the feedback loop. After a time equal to the packet length and guard times, the reset pulse is inserted into the flip flop unit. At that time the flip flop stop sending flip flop response at its output port. The flip flop window is then equal to the length of the data packet.

The optical AND logical gate has as input the flip flop’s output response that will be used then as a control signal and the data payload. After being launched at the input port1, the data packet at λ1 is split equally in two beams over the two MZI branches. Hence it arrives at both SOAs, reduces their carrier densities and modifies their optical gains almost with same quantity. The flip flop output is injected at input port3 it reaches only the SOA1. As a result an unbalanced gain and phase shifts of the two SOA-MZI branches is seen and the phase of the interference is altered. Consequently a data signal is obtained at the output port4 at wavelength λ1.

Accordingly, forwarding operation is attained when a data packet is fed into the gate along with a Set and a Reset pulse. The Set pulse is synchronized with the leading edge of the incoming data packet while the Reset pulse coincides with the end of the packet. The packets come out when the flip-flop is in ON state and blocked otherwise. Consequently the packet inserted with a matched set/reset is forwarded finally to the output port while others are blocked.

2.2 Contention resolution

The contention happens when two or more packets on different input ports attempt to go to the same output fiber port at the same time. Routers need to detect and resolve the contention according to the following three methods namely space domain (deflection routing), spectral domain (wavelength conversion) and time domain (time buffering).

In our case, the task is tougher since it consist in resolving the contention of variable length and asynchronous packets. The main issue will be then the conception of an all-optical router able to detect contending packet and send it to another output and which must operate autonomously and asynchronously. The buffering alternative needs variable delay lines which are complex to design and to control according to the packet length. The second alternative of spectral domain is investigated in this manuscript.

We start by reviewing the packet format used, that can accommodate for many potential cases as variable packet length and the use of more than one transmission rate across the network, etc. Packets are delimited on both beginning and end sides by set/reset pulses. Guard-times must be introduced between the signaling parts and the payload part of the packet.

We represent in Fig.2 the proposed router architecture with contention resolution based on wavelength conversion. It represents a 2×1 optical intermediate node. Our solution relies on combining packets detection. The packets are forwarded to output according FIFO (first in _first out) strategy of servicing. The scheme has 2 inputs (Input1 and Input2) and a single output (Output).

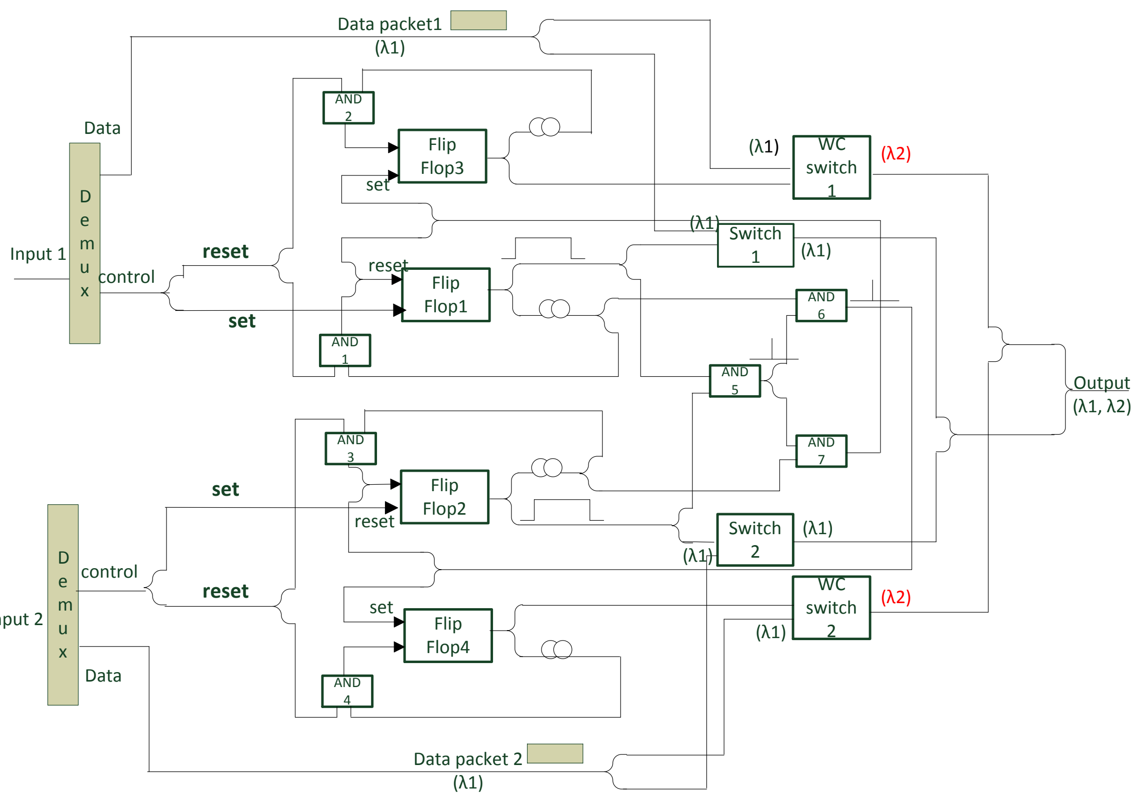

Figure 2: An all optical packet router architecture with contention resolution based on wavelength conversion strategy.

Two complementary operations are done. The first one is the packet forwarding and the second is the contention detection and resolution.

Packet forwarding is done thanks to packet forwarding blocks as described previsiouly in Fig.1. This work was validated in [7] for a single input and a single output. So in ordinary case where there is no contention, the packet arrives first at an input port (input1 or input 2). Then set/reset pulses and payload are extracted by a demultiplexer. The payload is then sanded to the forwarding unit whereas the extracted set/reset pulses are followed to the optical flip flop (flip flop1 or flip flop2 respectively) depending on the input port. The corresponding flip flop is then activated allowing the forward of packet to corresponding output port. The contention detection gate (AND5) does not detect a simultaneous presence of tow packets. Consequently no contention resolution treatment is done and the packet is forwarded through the AND gate switch (switch1 or switch 2 respectively) to the output port at wavelength (λ1).

However if a packet arrives at the other input port before the first packet is completely forwarded a contention occurs. In that case, contention needs to be detected as soon as it happens. Thus an AND operation between flip flop1 and flip flop2 responses is done by the AND5 logical gate. At that time we need to define the first coming packet. So we used two extra AND logical gates (AND6 and AND7 respectively). The AND6 logical gate is associated to the AND5 logical pulse and a delayed copy of flip flop1 window. Similarly the AND7 logical gate is associated to the AND5 logical pulse and a delayed copy of flip flop2 delayed window. So if the first coming packet is the packet at input1, AND6 will output a logical pulse else it is the AND7 that will send the logical pulse. This pulse is used to reset the flip flop (flip flop2 or flip flop1 respectively) and to set an extra flip flop (flip flop4 or flip flop3 respectively). First In First Out (FIFO) strategy is the adopted priority strategy for the contention resolution. A further wave conversion SOA-MZI forwarding gate is combined with this extra activated flip flop (flip flop4 or flip flop3) in order to forward the contending packet at a different wavelength ($\lambda 2$). Consequently the first packet will be forwarded to the appropriate output through the AND switch gate at the original wavelength ($\lambda 1$) while the second one will be forwarded at a different wavelength ($\lambda 2$).

Finally and to make sure that only activated flip flops are switched off a block of four optical logical AND gates are combined with the generated reset pulses.

The flip flop, the AND gates and the WC switches are implemented with a SOA-MZI configuration. These SOA-MZI configurations are the most promising candidate for their attractive features like the low energy requirement, simplicity, and compactness. Consequently in these gates and through the relation between the carrier density and the refractive index in the SOAs, the optical input signal can control the phase difference between the interferometer arms by means of cross-phase modulation (XPM).

The flip flop unit and the AND logical gate are described in Fig.1 while the wavelength conversion AND gate is described in the fig.3. In fact when we insert at input port 3 a data packets flow at wavelength $\lambda 1$ and we launch at input port 1 the flip flop output at wavelength $\lambda 2$ we got the AND logic operation as described previously and besides the output data is carried at wavelength $\lambda 2$. So we got at once the AND operation as well as the wave conversion in the same time.

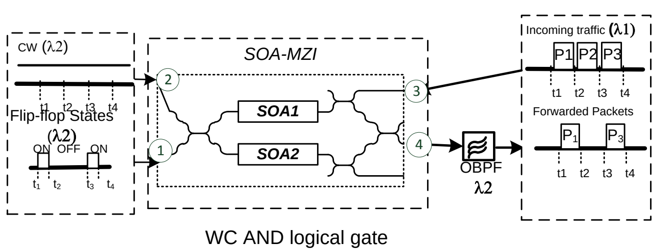

Figure 3: wavelength conversion AND logical gate.

3. RESULTS AND DISCUSSION

OPTIWAVE simulations are used to test the potentiality of the proposed router architecture to be employed as an intermediate packet routing node. The transmitter section consists on a laser diode that generates RZ 10Gbps pulse at wavelength 1545 nm. Optical time division multiplexed (OTDM) is used in order to get higher bit rates. Each transmitted bit of the RZ 10Gbps pulse is delayed at a particular time period and is then multiplexed and converted into a higher bit rate. Bit rate of 40Gbps and 100Gbps are obtained.

As shown in Fig.4 (a), (b) two variable length data packets flows at wavelength λ_1=1545nm are introduced from input 1 and input2 in order to reach the same output port. Contention occurs when two packet intent to reach the output port at the same time, so we have two case of contention in this figure test. When the first packet of the second flow is injected at input 2, the AND5 will detect the contention. Then the AND6 will detect that packet at input1 is the first coming hence the second coming packet at λ_1=1545nm is redirected to the contention resolution unit. After going through the wavelength conversion AND gate, the contending packet (packet at input port 2) is forwarded from the output port at λ_2=1555nm. So the two packets (from input 1 and from input2) are forwarded together from the same output port but at different wavelengths. The spectrum of forwarded packets at the output port is described in Fig.4 (e). The second introduced packet at input 1 and input 2 are forwarded from the output port at λ_2=1545nm since there is no contention. A second case of contention is detected for the third packets on the two data flows. In this time packet at input port 2 is the first coming. The same process is repeated but in this time packet at input port 1 is forwarded over the second wavelength λ_2=1555nm. Fig.4.(c) and fig.4.(d) show the output data packet at wavelength λ_1=1545nm and wavelength λ_2=1555nm respectively.

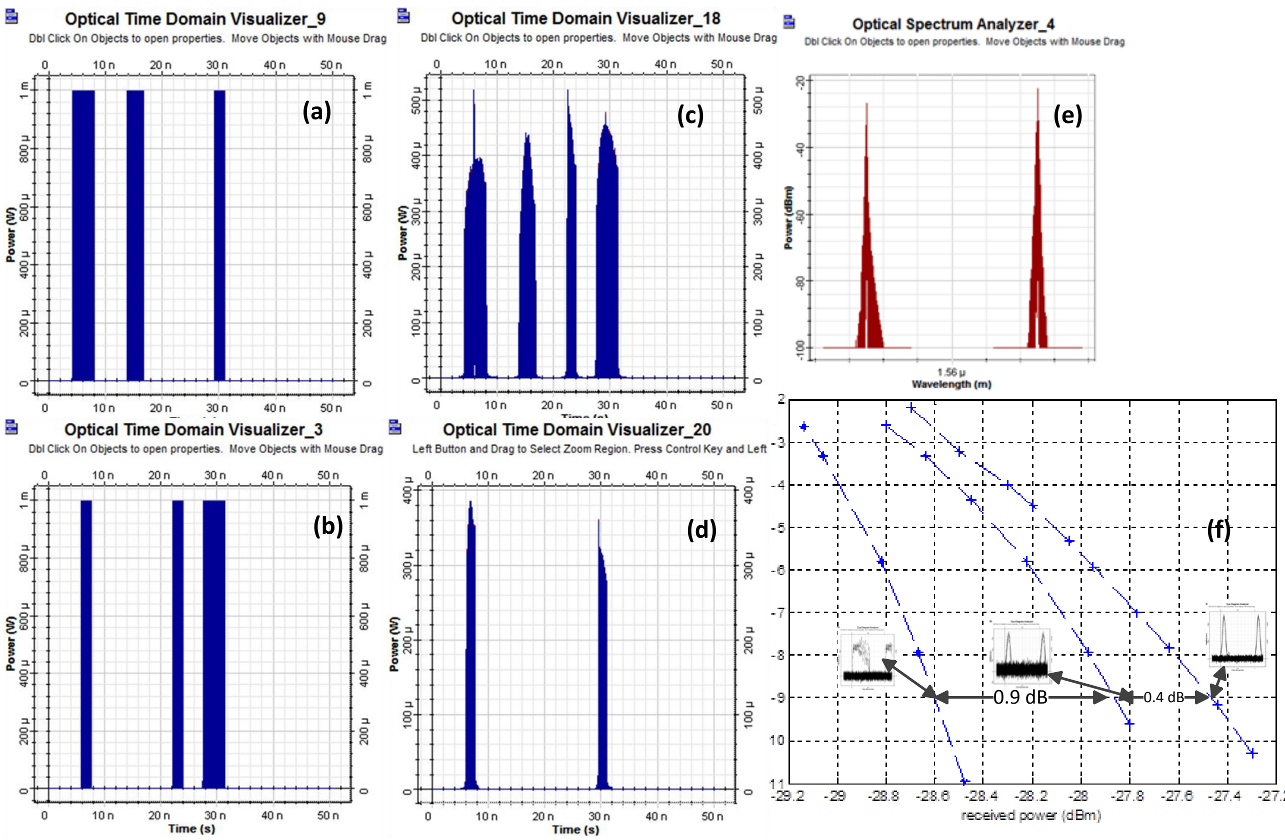

Figure 4:(a) input 1 data flow, (b) input 2 data flow, (c) output data at λ1 (d) output contending data packet at λ2, (e) output data spectrum, (f) BER curves of router based on wavelength conversion strategy 's forwrded packet.

As shown in Fig. 4 (f). Error free transmission, i.e. BER=10-9, is guaranteed for the proposed optical router based on wavelength conversion strategy for contention resolution at multiple bit rate (10 Gbps, 40 Gbps, and 100 Gbps). As described in Fig.5 the maximum power penalty for 40 Gbps in comparison with packet at 10 Gbps, is less than 1dB. While it is less of 0.5 dB for packet at 100 Gbps compared with packets at 40 Gbps. The eye diagram of output packets wavelength converted is well open and clean.

4. CONCLUSION

To deal with destination conflicts we have demonstrated a new all optical router architecture with contention resolution based on wavelength conversion for asynchronous variable length packets. So this architecture can forward two flows of data packets from two different inputs autonomously and asynchronously. FIFO (first in first out) strategy was the adopted strategy of servicing for forwarding packets in case of contention.

The proposed architectures can support bit rate up to 100 Gbps by using the optical time division multiplexed (OTDM) technology. The optical router with wavelength conversion contention resolution strategy was implemented with

Optiwave simulations. For forwarding operation and contention detection and resolution we exploits all optical flip flop and all optical AND gates responses. The technique is transparent to packet format BER measurements show error free operation for bit rates up to 100 Gbps.

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